

13.1 Construction of a Triangle**13.1.1 Construct a triangle having given:**

- Two sides and included angles.
- One side and two angles.
- Two of its sides and the angle opposite to one of them. With all three possibilities.

When two sides and the included angle are given.

Example Construct a triangle ABC in which
 $\overline{AB} = 4.8\text{ cm}$, $\overline{AC} = 3.2\text{ cm}$ and $m\angle B = 75^\circ$

Construction:

- Draw the line segment $\overline{AB}$ of measure 4.8 cm.
 - At point A, draw an angle $\angle XAB$ of measure 75° .
 - Cut $\overline{AC}$ of measure 3.2 cm from $\overrightarrow{AX}$
 - Draw $\overline{BC}$
- Thus $\triangle ABC$ is the required triangle.

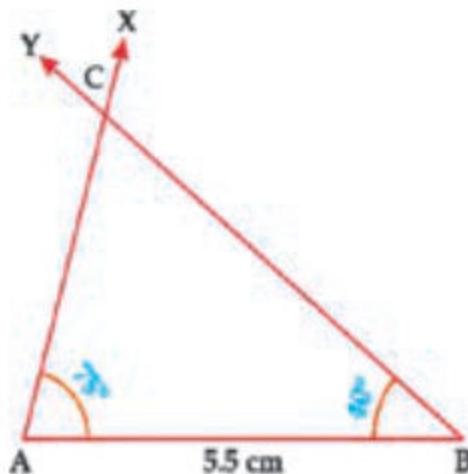


When one side and two angles are given.

Example 01 Construct a triangle $\triangle ABC$ in which
 $\overline{AB} = 5.5\text{ cm}$, $m\angle A = 75^\circ$ and $m\angle B = 40^\circ$

Construction:

- Draw the line segment $\overline{AB}$ of measure 5.5 cm.
 - At point A, draw an angle $\angle XAB$ of measure 75° .
 - At point B draw an angle $\angle YBA$ of measure 40° , such that $\overrightarrow{BY}$ cuts $\overrightarrow{AX}$ at point C.
- Thus $\triangle ABC$ is the required triangle.





Example 02 Construct a triangle $\triangle XYZ$ in which
 $m\angle A = 65^\circ$, $m\angle B = 40^\circ$ and $m\overline{BC} = 5.8$ cm.

Construction:

We know that in $\triangle ABC$

$$m\angle A + m\angle B + m\angle C = 180^\circ$$

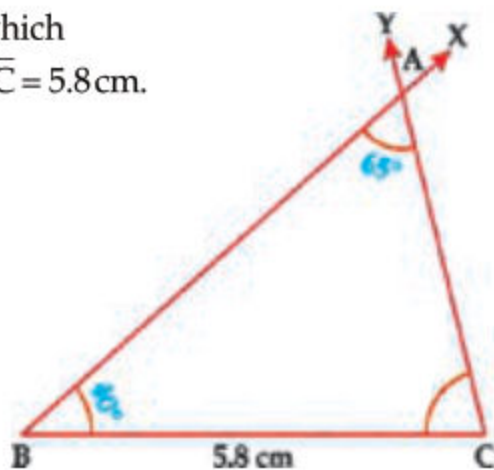
Here $m\angle A = 65^\circ$ and $m\angle B = 40^\circ$

$$\begin{aligned}\text{So, } m\angle C &= 180^\circ - (m\angle A + m\angle B) \\ &= 180^\circ - (65^\circ + 40^\circ) \\ &= 180^\circ - 105^\circ \\ &= 75^\circ\end{aligned}$$

We now construct the triangle with

$$m\overline{BC} = 5.8 \text{ cm, } m\angle B = 40^\circ \text{ and } m\angle C = 75^\circ$$

- Draw $\overline{BC}$ of measure 5.8 cm.
- Draw an angle $\angle XBC = 40^\circ$, at point B.
- Draw $m\angle YCB = 75^\circ$ at point C.
- Rays $\overrightarrow{BX}$ and $\overrightarrow{CY}$ intersect each other at point A,
Thus $\triangle ABC$ is the required triangle.



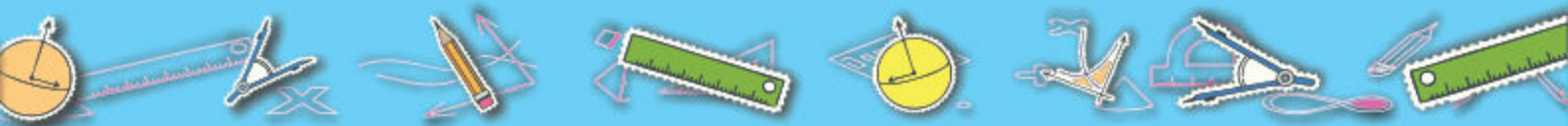
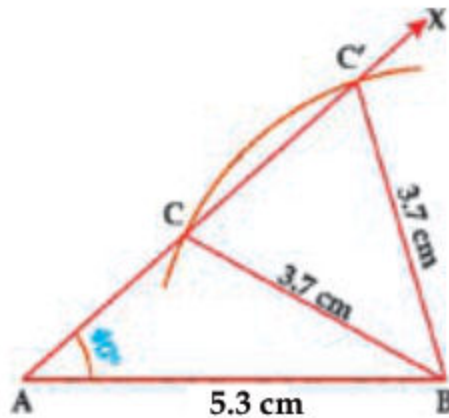
Two of its sides and the angle apposite to one of them. With all three possibilities.

Case I

Example 01 Construct a triangle ABC in which
 $m\angle A = 40^\circ$, $m\overline{BC} = 3.7$ cm and $m\overline{AB} = 5.3$ cm

Construction:

- Draw $\overline{AB}$ of measure 5.3 cm.
- At point A, draw $\angle BAX$ of measure 40° .
- With center B and radius 3.7 cm, draw an arc which cuts $\overrightarrow{AX}$ at point C and C' .
- Draw $\overline{BC}$ and $\overline{BC'}$
 $\triangle ABC$ and $\triangle ABC'$ are the required triangle.

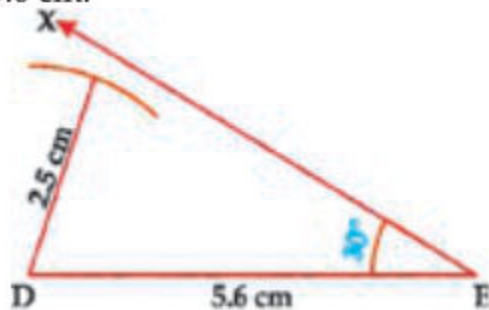


Case II

Example 02 Construct a triangle DEF when
 $m\overline{DE} = 5.6\text{ cm}$, $m\overline{DF} = 2.5\text{ cm}$ and $m\angle E = 30^\circ$

Construction:

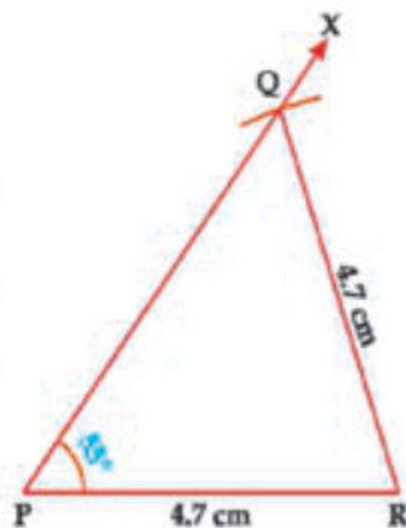
- Draw a line segment $\overline{DE}$ of measure 5.6 cm.
- Draw an angle DEX of measure 30° at point E.
- With D as a center draw an arc of radius 2.5 cm, which does not cut $\overrightarrow{EX}$ at any point. In this case no triangle can be constructed satisfying the given data.

**Case III**

Example 03 Construct a triangle PQR when
 $m\overline{PR} = m\overline{QR} = 4.7\text{ cm}$ and $m\angle P = 55^\circ$

Construction:

- Draw a line segment $\overline{PR}$ of measure 4.7 cm.
- Draw an angle $\angle XPR$ of measure 55° at point P.
- With point R as a center draw an arc of radius 4.7 cm, which cuts $\overrightarrow{PX}$ at point Q.
- Join point Q and R.
 $\triangle PQR$ is the required triangle.



Note: The above case I, case II and case III are called ambiguous cases.

Exercise 13.1

- Construct $\triangle PQR$ such that, $m\overline{PQ} = m\overline{QR} = 4.6\text{ cm}$ and $m\angle Q = 35^\circ$
- Construct $\triangle ABC$ such that, $m\overline{AB} = m\overline{AC} = 5.1\text{ cm}$ and $m\angle A = 65^\circ$
- Construct $\triangle LMN$ such that, $m\overline{LM} = 3.7\text{ cm}$, $m\overline{MN} = 2.5\text{ cm}$ and $\angle M = 50^\circ$
- Construct $\triangle ABC$ such that, $m\overline{AB} = 3.5\text{ cm}$, $m\overline{BC} = 2.7\text{ cm}$ and $\angle B = 110^\circ$
- Construct $\triangle XYZ$ such that, $m\overline{XY} = 4.1\text{ cm}$, $m\overline{YZ} = 5\text{ cm}$ and $\angle Z = 80^\circ$



6. Construct the $\triangle DEF$, $\triangle LMN$ and $\triangle ABC$ in the following.
 - (i) $m\overline{DE} = 5\text{cm}$, $m\angle D = 45^\circ$ and $m\angle E = 60^\circ$
 - (ii) $m\overline{LM} = 6\text{cm}$, $m\angle L = 75^\circ$ and $m\angle M = 45^\circ$
 - (iii) $m\overline{BC} = 5.8\text{cm}$, $m\angle A = 30^\circ$ and $m\angle B = 45^\circ$
7. Construct a $\triangle ABC$, when lengths of two of its sides and measure of an angle opposite one of the side is given as under:
 - (i) $m\overline{AC} = 4.5\text{cm}$, $m\overline{BC} = 4.1\text{cm}$ and $m\angle B = 75^\circ$
 - (ii) $m\overline{BC} = 5\text{cm}$, $m\overline{AB} = 5.5\text{cm}$ and $m\angle C = 70^\circ$
 - (iii) $m\overline{AB} = 5\text{cm}$, $m\overline{BC} = 5.5\text{cm}$ and $m\angle A = 45^\circ$

13.1.2 Draw

- Angle bisectors
- Altitudes
- Perpendicular bisectors
- Medians

(i) Draw the angle bisector of a given triangle

Example Draw bisectors of angle of $\triangle ABC$.

Given:

ABC is a triangle $\angle A$, $\angle B$ and $\angle C$ are its angles.

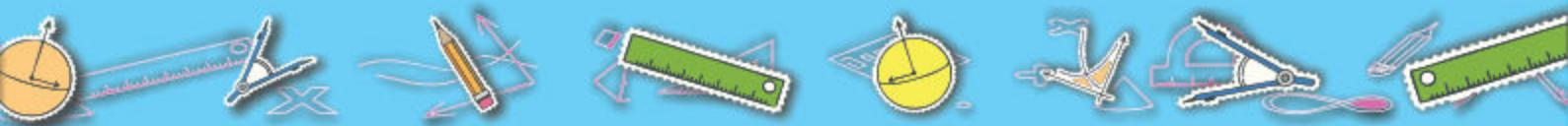
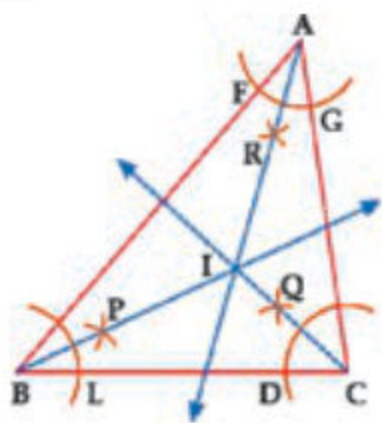
Required:

To draw bisectors of $\angle A$, $\angle B$ and $\angle C$.

Construction:

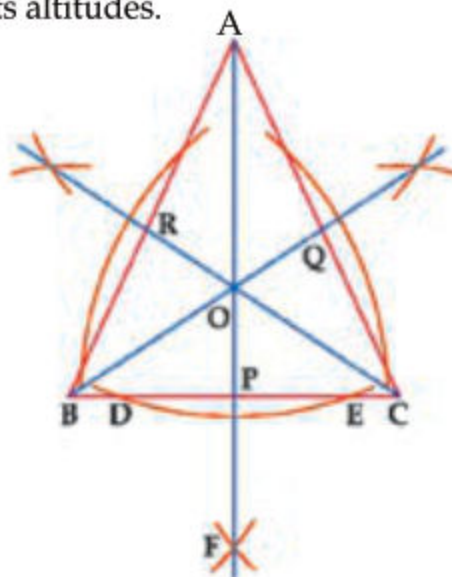
- i) Draw the triangle ABC .
- ii) With point B as a center draw an arc of any radius, intersecting the sides $\overline{BC}$ and $\overline{BA}$ at points L and M .
- iii) Take point L as a center and draw an arc of any radius.
- iv) Now take point M as a center and with the same radius draw another arc, which cuts the previous arc at point P .
- v) Join point P to B and produce it.
 $\overrightarrow{BP}$ is the bisector of $\angle B$.
- vi) Repeat steps (ii) to (v) to draw $\overrightarrow{CQ}$ and $\overrightarrow{AR}$ the bisectors of $\angle C$ and $\angle A$ respectively.

Hence, $\overrightarrow{BP}$, $\overrightarrow{CQ}$, and $\overrightarrow{AR}$ are the required bisector of $\triangle ABC$.



(ii) Draw the altitudes of a given triangle**Example** Take any triangle ABC and draw its altitudes.**Given:**A $\triangle ABC$ **Required:**To draw altitudes of the $\triangle ABC$.**Construction:**

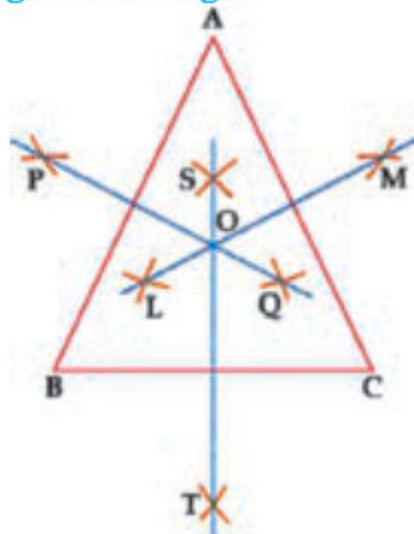
- Draw the triangle ABC.
- Take point A as center and draw an arc of suitable radius, which cuts $\overline{BC}$ at points D and E.
- From D as center, draw an arc of radius more than $\frac{1}{2}m\overline{DE}$.
- Again from point E draw another arc of same radius, cutting first arc at point F.
- Join the points A and F. Such that $\overline{AF}$ intersects $\overline{BC}$ at point P.
Then $\overline{AP}$ is the altitude of the $\triangle ABC$ from the vertex A.
- Repeat the steps (ii) to (v) and draw $\overline{BQ}$ and $\overline{CR}$, the altitudes of $\triangle ABC$ from the vertices B and C, respectively.
Hence $\overline{AP}$, $\overline{BQ}$ and $\overline{CR}$ are required altitudes of $\triangle ABC$

**(iii) Draw the perpendicular bisector of a given triangle****Example** Draw the perpendicular bisector of sides of a triangle ABC.**Given:**

A triangle ABC.

Required:To draw perpendicular bisectors of the sides $\overline{AB}$, $\overline{BC}$ and $\overline{CA}$.**Construction:**

- Draw the triangle ABC.
- To draw perpendicular bisector of the





- side $\overline{AB}$, with B as a center and radius more than half of $\overline{AB}$, draw arcs on either sides of $\overline{AB}$.
- iii) Now with A as a center and with the same radius, draw arcs on either sides of $\overline{AB}$, cutting previous arcs at P and Q.
- iv) Join P and Q.
 $\overline{PQ}$ is the perpendicular bisector of the $\overline{AB}$.
- v) Repeat the steps (ii) to (iv) and draw $\overline{ST}$ and $\overline{LM}$, the perpendicular bisectors of $\overline{BC}$ and $\overline{AC}$, respectively.
- Hence $\overline{PQ}$, $\overline{ST}$ and $\overline{LM}$ are the required perpendicular bisector of the sides $\overline{AB}$, $\overline{BC}$ and $\overline{AC}$, respectively, of the $\triangle ABC$.

(iv) Draw the median of a given triangle

Example Take any triangle ABC and draw medians of this triangle.

Given:

A $\triangle ABC$

Required:

To draw medians of the $\triangle ABC$.

Construction:

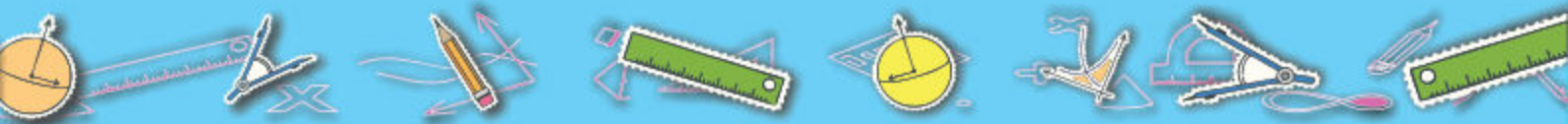
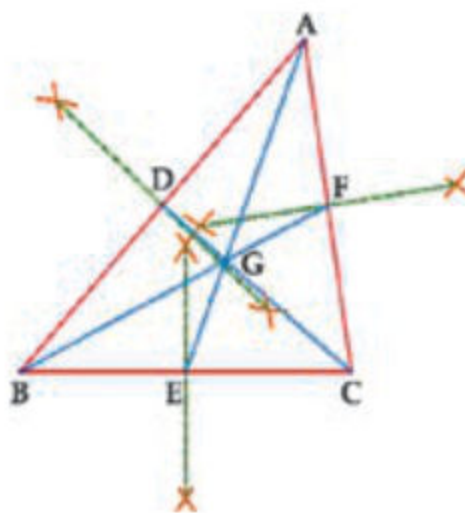
- Draw the triangle ABC.
- Bisect the sides $\overline{AB}$, $\overline{BC}$ and $\overline{AC}$ at points D, E and F, respectively.
- Join A to E; B to F and C to D.

Thus $\overline{AE}$, $\overline{BF}$ and $\overline{CD}$ are the required medians of the $\triangle ABC$, which meet in a point G.

It may be noted that medians of every triangle are concurrent (i.e., meet in one point) and their point of concurrency, called centroid, divides each of them in 2 : 1.

By actual measurement it can be proved that

$$\frac{m\overline{AG}}{m\overline{GE}} = \frac{m\overline{BG}}{m\overline{GF}} = \frac{m\overline{CG}}{m\overline{GD}} = \frac{2}{1}$$



Exercise 13.2

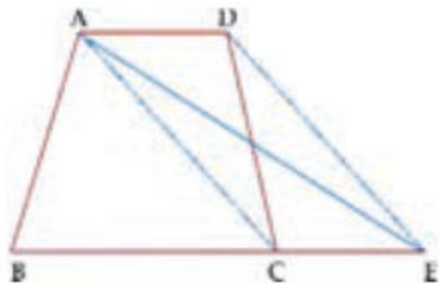
1. Take a Δ and draw the medians and prove that they are concurrent.
2. Take a Δ and draw the altitudes and prove that they are concurrent.
3. Take a Δ and draw the internal bisectors of angles and prove that they are concurrent.
4. Construct a triangle ABC in which $m\overline{BC} = 6$ cm, $m\overline{CA} = 4$ cm and $m\overline{AB} = 5$ cm, draw the bisectors of angles A and B.
5. Construct a triangle PQR in which $m\overline{PQ} = 5.7$ cm, $m\overline{QR} = 6.4$ cm and $m\overline{PR} = 4.4$ cm, draw the altitudes from vertex R and vertex Q.
6. Construct a triangle STU in which $\angle T = 60^\circ$, $\angle U = 30^\circ$ and $m\overline{TU} = 7$ cm. Find the perpendicular bisectors of the sides of triangle and prove that they are concurrent.
7. Construct a right triangle ABC in which $\angle C = 90^\circ$, $\angle B = 45^\circ$ and $m\overline{CB} = 5$ cm. Draw the medians of the triangle.
8. Construct the following ΔXYZ . Draw their three medians and show that they are concurrent.
 - (i) $m\overline{YZ} = 4.4$ cm, $m\angle Y = 45^\circ$ and $m\angle Z = 75^\circ$
 - (ii) $m\overline{XY} = 4.6$ cm, $m\overline{XZ} = 4.6$ cm and $m\angle Y = 60^\circ$
9. Construct the ΔKLM , in which $m\overline{KL} = 4.8$ cm, $m\overline{LM} = 5.2$ cm and $m\overline{MK} = 4.5$ cm, draw their altitudes and verify their concurrency.
10. Construct the ΔPQR , in which $m\overline{PQ} = 7$ cm, $m\overline{QR} = 6.5$ cm and $m\overline{PR} = 5.8$ cm, find their perpendicular bisectors and verify their concurrency.

13.2 Figures with equal Areas

13.2.1 Construct a triangle equal in area to a given quadrilateral.

E.g. draw a triangle equal in area to given quadrilateral ABCD. We know that, Area of all triangles with same base equal of vertices are on the line perpendicular to base.

1. ABCD is a given quadrilateral.
2. Join A to C.
4. Through D, draw $\overline{DE}$ parallel to $\overline{AC}$ meeting $\overline{BC}$ produced at point E.
5. Join A to E, then ABE is the required triangle.

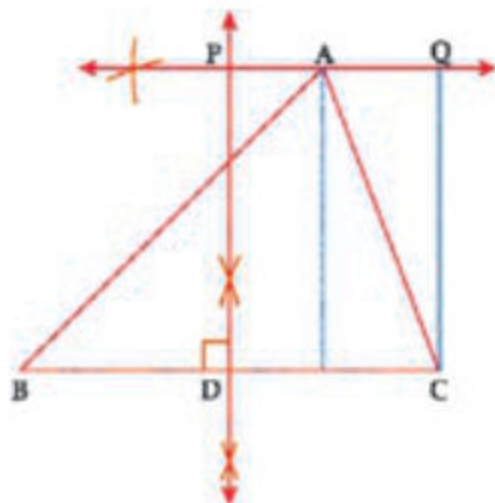




13.2.2 Construct a rectangle equal in area to a given triangle.

E.g. Construct a rectangle equal in area to given $\triangle ABC$

1. Draw a triangle ABC.
2. Draw a perpendicular bisector $\overleftrightarrow{PD}$ of $\overline{BC}$.
3. Through A, draw a line PQ parallel to $\overline{BC}$.
4. Take $m\overline{PQ} = m\overline{DC}$.
5. Then CDPQ is the required rectangle.

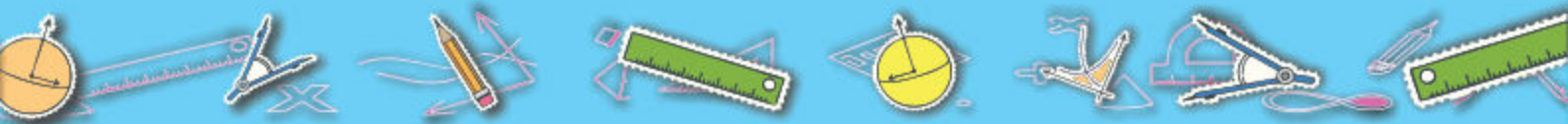
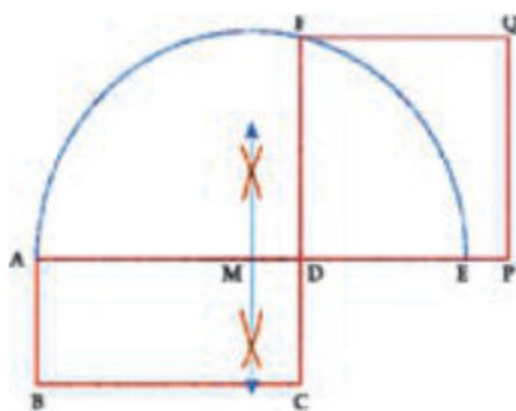


13.2.3 Construct a Square equal in area to a given rectangle.

E.g. Construct a square equal in Area to given rectangle ABCD.

Following are the steps of construction

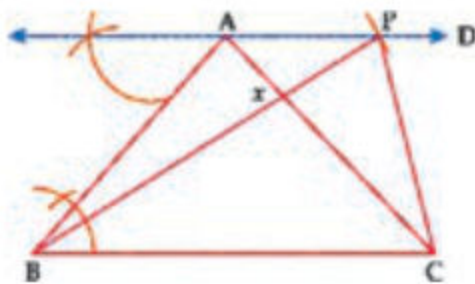
1. ABCD is a given rectangle.
2. Produce side $\overline{AD}$ to E making $m\overline{DE} = m\overline{CD}$.
3. Bisect $\overline{AE}$ at M.
4. With centre M and radius $m\overline{AM}$ construct a semi circle.
5. Produce $\overline{CD}$ to meet the semi circle at F.
6. On $\overline{DF}$ as a side construct a square DFQP. This shall be required square.



13.2.4 Construct a Triangle of equivalent area on a base of given length:

Following are the steps of construction

1. ABC is given triangle.
2. Draw $\overleftrightarrow{AD} \parallel \overleftrightarrow{BC}$.
3. With B as centre, and radius = x , such that $m\overline{BC} = x$ draw an arc cutting $\overleftrightarrow{AD}$ at P.
4. Join $\overline{BP}$ and $\overline{CP}$.
5. Then $\triangle BCP$ is the required triangle with equal base $\overline{BP} = x$ and area equivalent to $\triangle ABC$.



Exercise 13.3

1. Construct a rectangle whose adjacent sides are 2.5 cm and 5 cm respectively. Construct a square having area equal to the given rectangle.
2. Construct a square equal in area to a rectangle whose adjacent sides are 4.5 cm and 2.2 cm respectively. Measure the sides of the square and find its area and compare with the area of the rectangle. Also verify by measurement that the perimeter of the square is less than that of the rectangle.
3. Construct a triangle having base 4 cm and other two sides equal to 3.6 cm and 3.8 cm each. Transform it into a rectangle with equal Area.
4. Construct a triangle having base 6 cm and other sides equal to 5 cm and 6 cm each. Construct a rectangle equal in area to given Δ .

Review Exercise 13

1. Fill in the blanks.
 - i) The side of a right triangle opposite to the right angle is _____
 - ii) The line segment joining a vertex of a triangle and perpendicular to its opposite side is called an _____
 - iii) A line segment drawn from a vertex of a triangle and meeting the mid-point of its opposite side is called a _____



- iv) The perpendicular bisectors of the sides of a triangle are _____
v) Two or more than two triangles are said to be congruent if they are equiangular and measures of their corresponding sides are _____

2. Tick (✓) the correct answer.

- i) A triangle having all the three sides congruent is called _____ triangle.

(a) scalene (b) right angled
(c) equilateral (d) isosceles

- ii) A quadrilateral having each angle equal to 90° and all the sides are congruent is called _____

(a) parallelogram (b) rectangle
(c) trapezium (d) square

- iii) The medians of a triangle are _____

(a) collinear (b) congruent
(c) concurrent (d) parallel

- iv) The _____ altitudes of an equilateral triangle are congruent.

(a) two (b) three
(c) four (d) none

- v) The diagonals of a rectangle _____ each other.

(a) bisect (b) trisect
(c) bisect at right angle (d) none of these

- vi) The _____ of a triangle cut each other in the ratio 2:1.

(a) Altitudes (b) Angle bisectors
(c) Right bisectors (d) Medians

- vii) If each angle on the base of an isosceles triangle is 45° , then the measure of the third angle is _____

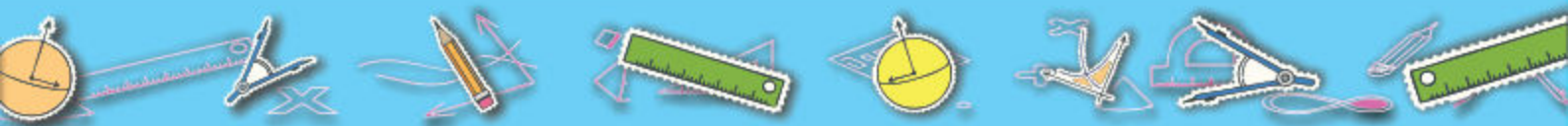
(a) 30° (b) 60°
(c) 90° (d) 120°

- viii) If the three medians of a triangle are congruent then the triangle is _____

(a) Right angled (b) equilateral
(c) Isosceles (d) acute angled

- ix) If two _____ of a triangle are congruent, then the triangle will be isosceles.

(a) Altitudes (b) Medians
(c) Right bisectors (d) sides





Summary

- ◆ In this unit we have learnt the construction of the following figures and relevant concepts.
- ◆ To construct a triangle, having given two sides and the included angle.
- ◆ To construct a triangle, having given one side and two of the angles.
- ◆ To construct a triangle, having given two of its sides and the angle opposite to one of them.
- ◆ To draw angle bisectors of a given triangle and to verify their concurrency.
- ◆ To draw altitudes of a given triangle and verify their concurrency.
- ◆ To draw perpendicular bisectors of the sides of a given triangle and to verify their concurrency.
- ◆ To draw medians of a given triangle and verify their concurrency.
- ◆ To construct a triangle equal in area to a given quadrilateral.
- ◆ To construct a rectangle equal in area to given triangle.
- ◆ To construct a square equal in area to given rectangle.
- ◆ To construct a triangle of equivalent area on the base of given length.
- ◆ Two or more than three lines are said to be concurrent if these pass through a common point and that point is called the point of concurrency.
- ◆ The point where the internal bisectors of the angles of a triangle intersect is called the in-centre of a triangle.
- ◆ The point of concurrency of the perpendicular bisectors of the sides of a triangle is called its circum-centre.
- ◆ Median of a triangle means a line segment joining a vertex of a triangle to the mid-point of the opposite side.
- ◆ Ortho-centre of a triangle means the point of concurrency of three altitudes of a triangle.

Unit

14

• Weightage = 5%

THEOREMS RELATED WITH AREA

Student Learning Outcomes (SLOs)

After completing this unit, students will be able to:

- ◆ Understand the following theorems along with their corollaries and apply them to solve allied problems.
- ◆ Parallelograms on the same base and lying between the same parallel lines (or of the same altitude) are equal in area.
- ◆ Parallelograms on equal bases and having the same altitude are equal in area.
- ◆ The triangles on the same base and of the same altitude are equal in area.
- ◆ Triangles on equal bases and of the same altitude are equal in area.